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BDA Practical 1

Problem Statement: Set up and Configuration Hadoop Using CloudEra/ Google Cloud BigQuery. Databricks Lakehouse Platform. Snowflake. Amazon Redshift

Step 1: Install Java

sudo apt install openjdk-8-jdk -y

```
vboxuser@Ubuntu-24:~$ sudo apt install openjdk-8-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-8-jdk is already the newest version (8u482-ga-us1-0ubuntu1~24.04).
0 upgraded, 0 newly installed, 0 to remove and 9 not upgraded.
vboxuser@Ubuntu-24:~$
```

Step 2: Set JAVA_HOME

nano ~/.bashrc

```
vboxuser@Ubuntu-24:~$ nano ~/.bashrc
vboxuser@Ubuntu-24:~$ source ~/.bashrc
```

Add:

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

export HADOOP_HOME=~/.hadoop

export PATH=\$HADOOP_HOME/bin:\$HADOOP_HOME/sbin:\$PATH

source ~/.bashrc

```
GNU nano 7.2 /home/vboxuser/.bashrc
if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
    if [ -f /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -f /etc/bash_completion ]; then
        . /etc/bash_completion
    fi
fi

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_HOME=~/.hadoop
export PATH=$HADOOP_HOME/bin:$HADOOP_HOME/sbin:$PATH
```

Step 3: Download and Extract Hadoop

wget https://downloads.apache.org/hadoop/common/hadoop-3.4.3/hadoop-3.4.3.tar.gz

```
vboxuser@Ubuntu-24:~$ wget https://downloads.apache.org/hadoop/common/hadoop-3.4.3/hadoop-3.4.3.tar.gz
--2026-03-18 17:26:24-- https://downloads.apache.org/hadoop/common/hadoop-3.4.3/hadoop-3.4.3.tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f9:3a:2c57::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|135.181.214.104|:443..
. connected.
HTTP request sent, awaiting response... 200 OK
Length: 514917037 (491M) [application/x-gzip]
Saving to: 'hadoop-3.4.3.tar.gz'

hadoop-3.4.3.tar.gz 100%[=====>] 491.06M  3.37MB/s   in 3m 7s

2026-03-18 17:29:31 (2.63 MB/s) - 'hadoop-3.4.3.tar.gz' saved [514917037/514917037]
```

tar -xvzf hadoop-3.4.3.tar.gz

mv hadoop-3.4.3 hadoop

```
vboxuser@Ubuntu-24:~$ tar -xvzf hadoop-3.4.3.tar.gz
```

```
hadoop-3.4.3/etc/hadoop/yarn-env.sh
hadoop-3.4.3/etc/hadoop/yarn-env.cmd
hadoop-3.4.3/etc/hadoop/workers
hadoop-3.4.3/etc/hadoop/mapred-queues.xml.template
hadoop-3.4.3/etc/hadoop/shellprofile.d/
hadoop-3.4.3/etc/hadoop/shellprofile.d/example.sh
hadoop-3.4.3/etc/hadoop/hadoop-user-functions.sh.example
hadoop-3.4.3/etc/hadoop/user_ec_policies.xml.template
hadoop-3.4.3/etc/hadoop/mapred-env.sh
hadoop-3.4.3/etc/hadoop/kms-acls.xml
hadoop-3.4.3/etc/hadoop/mapred-site.xml
hadoop-3.4.3/etc/hadoop/httpfs-env.sh
hadoop-3.4.3/etc/hadoop/log4j.properties
hadoop-3.4.3/etc/hadoop/hadoop-env.cmd
hadoop-3.4.3/etc/hadoop/kms-env.sh
hadoop-3.4.3/etc/hadoop/hdfs-rbf-site.xml
hadoop-3.4.3/etc/hadoop/kms-log4j.properties
hadoop-3.4.3/etc/hadoop/hadoop-env.sh
hadoop-3.4.3/etc/hadoop/configuration.xsl
hadoop-3.4.3/etc/hadoop/kms-site.xml
hadoop-3.4.3/etc/hadoop/mapred-env.cmd
hadoop-3.4.3/etc/hadoop/container-executor.cfg
vboxuser@Ubuntu-24:~$ mv hadoop-3.4.3 hadoop
```

Step 4: Configure Hadoop Environment

nano ~/hadoop/etc/hadoop/hadoop-env.sh

```
vboxuser@Ubuntu-24:~$ nano ~/hadoop/etc/hadoop/hadoop-env.sh
```

Add:

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

```
GNU nano 7.2 /home/vboxuser/hadoop/etc/hadoop/hadoop-env.sh *
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
# Licensed to the Apache Software Foundation (ASF) under one
# or more contributor license agreements. See the NOTICE file
# distributed with this work for additional information
# regarding copyright ownership. The ASF licenses this file
# to you under the Apache License, Version 2.0 (the
# "License"); you may not use this file except in compliance
# with the License. You may obtain a copy of the License at
#
```

Step 5: Configure core-site.xml

nano ~/hadoop/etc/hadoop/core-site.xml

```
vboxuser@Ubuntu-24:~$ nano ~/hadoop/etc/hadoop/core-site.xml
vboxuser@Ubuntu-24:~$ nano ~/hadoop/etc/hadoop/hdfs-site.xml
```

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

```

GNU nano 7.2 /home/vboxuser/hadoop/etc/hadoop/core-site.xml
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http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>

```

Step 6: Configure hdfs-site.xml

nano ~/hadoop/etc/hadoop/hdfs-site.xml

```

vboxuser@Ubuntu-24:~$ nano ~/hadoop/etc/hadoop/core-site.xml
vboxuser@Ubuntu-24:~$ nano ~/hadoop/etc/hadoop/hdfs-site.xml

```

```

<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///home/vboxuser/hadoopdata/namenode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///home/vboxuser/hadoopdata/datanode</value>
  </property>
</configuration>

```

```

GNU nano 7.2 /home/vboxuser/hadoop/etc/hadoop/hdfs-site.xml *
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///home/vboxuser/hadoopdata/namenode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///home/vboxuser/hadoopdata/datanode</value>
  </property>
</configuration>

```

Step 7: Setup SSH

ssh-keygen -t rsa -P ""

cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys

chmod 0600 ~/.ssh/authorized_keys

ssh localhost

```
vboxuser@Ubuntu-24:~$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/vboxuser/.ssh/id_rsa):
/home/vboxuser/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Your identification has been saved in /home/vboxuser/.ssh/id_rsa
Your public key has been saved in /home/vboxuser/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:kVH1bwFddu/wbM0Vsky48golbZNLU0dj27sql6a3kA vboxuser@Ubuntu-24
The key's randomart image is:
+---[RSA 3072]---+
|      . . . . .+|
|      o. +o.*+|
|      o+ o *++|
|      o.* ++=.o|
|      .SoE+..+B+|
|      ..  +oo*|
|      . . oo |
|      * . |
|      o*.o |
+---[SHA256]---+
```

```
vboxuser@Ubuntu-24:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
vboxuser@Ubuntu-24:~$ chmod 0600 ~/.ssh/authorized_keys
vboxuser@Ubuntu-24:~$ ssh localhost
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-19-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

2 additional security updates can be applied with ESM Apps.
```

Step 8: Create Hadoop Directories

```
mkdir -p ~/hadoopdata/namenode
```

```
mkdir -p ~/hadoopdata/datanode
```

```
vboxuser@Ubuntu-24:~$ mkdir -p ~/hadoopdata/namenode
vboxuser@Ubuntu-24:~$ mkdir -p ~/hadoopdata/datanode
vboxuser@Ubuntu-24:~$
```

Step 9: Format NameNode

```
hdfs namenode -format
```

```
vboxuser@Ubuntu-24:~$ hdfs namenode -format
2026-03-18 17:46:12,675 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = Ubuntu-24/10.0.2.15
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.4.3
STARTUP_MSG: classpath = /home/vboxuser/hadoop/etc/hadoop:/home/vboxuser/hadoop/share
/hadoop/common/lib/jsr311-api-1.1.1.jar:/home/vboxuser/hadoop/share/hadoop/common/lib/c
ommons-net-3.9.0.jar:/home/vboxuser/hadoop/share/hadoop/common/lib/nimbus-jose-jwt-10.4
.jar:/home/vboxuser/hadoop/share/hadoop/common/lib/netty-codec-mqtt-4.1.127.Final.jar:/
home/vboxuser/hadoop/share/hadoop/common/lib/jetty-server-9.4.57.v20241219.jar:/home/vb
oxuser/hadoop/share/hadoop/common/lib/jakarta.activation-api-1.2.1.jar:/home/vboxuser/h
adoop/share/hadoop/common/lib/snappy-java-1.1.10.4.jar:/home/vboxuser/hadoop/share/hado
op/common/lib/netty-transport-native-kqueue-4.1.127.Final-osx-x86_64.jar:/home/vboxuser
/hadoop/share/hadoop/common/lib/jetty-xml-9.4.57.v20241219.jar:/home/vboxuser/hadoop/sh
are/hadoop/common/lib/jsr305-3.0.2.jar:/home/vboxuser/hadoop/share/hadoop/common/lib/ne
tty-transport-native-epoll-4.1.127.Final-linux-aarch_64.jar:/home/vboxuser/hadoop/share
/hadoop/common/lib/jackson-core-2.12.7.jar:/home/vboxuser/hadoop/share/hadoop/common/li
b/javax.servlet-api-3.1.0.jar:/home/vboxuser/hadoop/share/hadoop/common/lib/netty-codec
*****/
```

Step 10: Start Hadoop

```
start-dfs.sh
```

```
jps
```

```
vboxuser@Ubuntu-24:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Ubuntu-24]
vboxuser@Ubuntu-24:~$ jps
10177 SecondaryNameNode
9943 DataNode
10296 Jps
9807 NameNode
vboxuser@Ubuntu-24:~$
```

Step 11: Create Input File

echo "hello hadoop hello world" > file1.txt

```
vboxuser@Ubuntu-24:~$ echo "hello hadoop hello world" > file1.txt
vboxuser@Ubuntu-24:~$
```

Step 12: Create HDFS Directory and Upload File to HDFS

```
hdfs dfs -mkdir /input
```

```
hdfs dfs -put file1.txt /input
```

```
vboxuser@Ubuntu-24:~$ hdfs dfs -mkdir /input
vboxuser@Ubuntu-24:~$ hdfs dfs -put file1.txt /input
vboxuser@Ubuntu-24:~$
```

Step 13: Run WordCount Program

```
hadoop jar ~/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.4.3.jar wordcount /input /output
```

```
vboxuser@Ubuntu-24:~$ hadoop jar ~/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.4.3.jar wordcount /input /output
2026-03-18 17:49:26,752 INFO input.FileInputFormat: Total input files to process : 1
2026-03-18 17:49:26,775 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-18 17:49:26,908 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1825583043_0001
2026-03-18 17:49:26,912 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-18 17:49:27,057 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-03-18 17:49:27,058 INFO mapreduce.Job: Running job: job_local1825583043_0001
2026-03-18 17:49:27,058 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-18 17:49:27,067 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-03-18 17:49:27,069 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-18 17:49:27,070 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-03-18 17:49:27,070 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-03-18 17:49:27,128 INFO mapred.LocalJobRunner: Waiting for map tasks
```

Step 15: View Output

```
hdfs dfs -cat /output/part-r-00000
```

```
vboxuser@Ubuntu-24:~$ hdfs dfs -cat /output/part-r-00000
hadoop 1
hello 2
world 1
vboxuser@Ubuntu-24:~$
```

Result:

Hadoop was successfully installed and configured in pseudo-distributed mode. The WordCount MapReduce program was executed successfully and output was obtained.

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BDA Practical 2

Problem Statement: To Develop a MapReduce program to calculate the frequency of a given word in a given file.

STEP 1: Create Java File and Write Code

nano WordCount.java

```
vboxuser@Ubuntu-24:~/Desktop$ nano WordCount.java
```

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCount {
    // Mapper
    public static class MyMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
        public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {
            String[] words = value.toString().split(" ");
            for (String word : words) {
                context.write(new Text(word.toUpperCase().trim()), new IntWritable(1));
            }
        }
    }

    // Reducer
    public static class MyReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
        public void reduce(Text key, Iterable<IntWritable> values, Context context)
            throws IOException, InterruptedException {
            int sum = 0;
            for (IntWritable val : values) {
                sum += val.get();
            }
            context.write(key, new IntWritable(sum));
        }
    }

    // Main
    public static void main(String[] args) throws Exception {

        Job job = Job.getInstance(new Configuration(), "WordCount");
```

```

    job.setJarByClass(WordCount.class);
    job.setMapperClass(MyMapper.class);
    job.setReducerClass(MyReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

STEP 2: Compile (Java 8 compatibility)

javac -source 1.8 -target 1.8 -classpath `hadoop classpath` -d . WordCount.java

```

vboxuser@Ubuntu-24:~/Desktop$ javac -source 1.8 -target 1.8 -classpath `hadoop classpath` -d . WordCount.java
warning: [options] bootstrap class path not set in conjunction with -source 8
warning: [options] source value 8 is obsolete and will be removed in a future release
warning: [options] target value 8 is obsolete and will be removed in a future release
warning: [options] To suppress warnings about obsolete options, use -Xlint:-options.
4 warnings
vboxuser@Ubuntu-24:~/Desktop$

```

STEP 3: Create JAR file

jar -cvf wordcount.jar *.class

```

vboxuser@Ubuntu-24:~/Desktop$ jar -cvf wordcount.jar *.class
added manifest
adding: WordCount$MyMapper.class(in = 1770) (out= 748)(deflated 57%)
adding: WordCount$MyReducer.class(in = 1668) (out= 715)(deflated 57%)
adding: WordCount.class(in = 1402) (out= 774)(deflated 44%)
vboxuser@Ubuntu-24:~/Desktop$

```

STEP 4: Start Hadoop

start-dfs.sh

start-yarn.sh

```

vboxuser@Ubuntu-24:~/Desktop$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Ubuntu-24]
vboxuser@Ubuntu-24:~/Desktop$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
vboxuser@Ubuntu-24:~/Desktop$ jps
4341 NodeManager
3958 SecondaryNameNode
4487 Jps
3768 DataNode
3644 NameNode
4221 ResourceManager

```

STEP 5: Create Input File

nano wordcount.txt

Bus, Car, bus, car, train, car, bus, BUS


```
vboxuser@Ubuntu-24:~/Desktop$ nano wordcount.txt
vboxuser@Ubuntu-24:~/Desktop$
```

STEP 6: Create HDFS Input Folder and Upload File to HDFS

```
hadoop fs -mkdir /input
```

```
hadoop fs -put wordcount.txt /input
```

```
vboxuser@Ubuntu-24:~/Desktop$ hadoop fs -mkdir /input
vboxuser@Ubuntu-24:~/Desktop$ hadoop fs -put wordcount.txt /input
vboxuser@Ubuntu-24:~/Desktop$
```

STEP 7: Run MapReduce Job

```
hadoop jar wordcount.jar WordCount /input /output
```

```
vboxuser@Ubuntu-24:~/Desktop$ hadoop jar wordcount.jar WordCount /input /output
2026-03-19 17:09:38,527 WARN mapreduce.JobResourceUploader: Hadoop command-line option
parsing not performed. Implement the Tool interface and execute your application with T
oolRunner to remedy this.
2026-03-19 17:09:38,926 INFO input.FileInputFormat: Total input files to process : 1
2026-03-19 17:09:39,011 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-19 17:09:39,364 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_loc
al1237481832_0001
2026-03-19 17:09:39,370 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-19 17:09:39,931 INFO mapreduce.Job: The url to track the job: http://localhost:
8080/
2026-03-19 17:09:39,937 INFO mapreduce.Job: Running job: job_local1237481832_0001
2026-03-19 17:09:39,950 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-19 17:09:40,007 INFO output.PathOutputCommitterFactory: No output committer fac
tory defined, defaulting to FileOutputCommitterFactory
2026-03-19 17:09:40,014 INFO output.FileOutputCommitter: File Output Committer Algorith
m version is 2
2026-03-19 17:09:40,014 INFO output.FileOutputCommitter: FileOutputCommitter skip clean
up _temporary folders under output directory:false, ignore cleanup failures: false
2026-03-19 17:09:40,020 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop
p.mapreduce.lib.output.FileOutputCommitter
```

STEP 8: Check Output

```
hadoop fs -cat /output/part-r-00000
```

```
vboxuser@Ubuntu-24:~/Desktop$ hadoop fs -cat /output/part-r-00000
BUS      4
CAR      3
TRAIN   1
vboxuser@Ubuntu-24:~/Desktop$
```

DR. D. Y. PATIL INSTITUTE OF ENGINEERING, MANAGEMENT AND RESEARCH Akurdi, Pune-411044 Department-Artificial Intelligence and Data Science		
Laboratory Journal Subject: Computer Laboratory – IV		AY:2025-26
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Lab Practical 3

To Develop a Map Reduce program to implement Matrix Multiplication.

Step 1: Create Working Directory

Commands:

```
mkdir ~/matrixmultiply
cd ~/matrixmultiply
```

Step 2: Create Matrix M Input File

Command:

```
nano M.txt
```

Enter the following content (row, column, value):

```
0,0,1
0,1,2
0,2,3
1,0,4
1,1,5
1,2,6
```

Step 3: Create Matrix N Input File

Command:

```
nano N.txt
```

Enter the following content (row, column, value):

```
0,0,7
0,1,8
1,0,9
1,1,10
2,0,11
2,1,12
```

Step 4: Create Combined Input File

Command:

```
nano input.txt
```

Enter the following content (matrix name, row, column, value):

```

athy@HP:~/matrixmultiply$ cat input.txt
M,0,0,1
M,0,1,2
M,0,2,3
M,1,0,4
M,1,1,5
M,1,2,6
N,0,0,7
N,0,1,8
N,1,0,9
N,1,1,10
N,2,0,11
N,2,1,12

```

Step 5: Create the Java Program

Command:

```
nano Multiply.java
```

```

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

```

```

public class Multiply {
    public static class MatrixMapper extends
        Mapper<LongWritable, Text, Text, Text> {
        public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {
            String line = value.toString();
            String[] tokens = line.split(",");
            if (tokens.length == 4) {
                String matrix = tokens[0];
                int i = Integer.parseInt(tokens[1]);
                int j = Integer.parseInt(tokens[2]);
                double val = Double.parseDouble(tokens[3]);
                if (matrix.equals("M")) {
                    for (int k = 0; k < 2; k++) {
                        context.write(new Text(i+"", "+k),
                            new Text("M, "+j+"", "+val));
                    }
                } else {
                    for (int r = 0; r < 2; r++) {
                        context.write(new Text(r+"", "+j),
                            new Text("N, "+i+"", "+val));
                    }
                }
            }
        }
    }
}

```

```

    }
  }
}

public static class MatrixReducer extends
    Reducer<Text, Text, Text, Text> {
    public void reduce(Text key, Iterable<Text> values,
        Context context)
        throws IOException, InterruptedException {
        double[] M = new double[100];
        double[] N = new double[100];
        for (Text val : values) {
            String[] parts = val.toString().split(",");
            int j = Integer.parseInt(parts[1]);
            double v = Double.parseDouble(parts[2]);
            if (parts[0].equals("M")) { M[j] = v; }
            else { N[j] = v; }
        }
        double sum = 0;
        for (int j = 0; j < 100; j++) {
            sum += M[j] * N[j];
        }
        context.write(key, new Text(String.valueOf(sum)));
    }
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "MatrixMultiply");
    job.setJarByClass(Multiply.class);
    job.setMapperClass(MatrixMapper.class);
    job.setReducerClass(MatrixReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

Step 6: Compile the Java Program

Command:

```
javac -classpath $(hadoop classpath) Multiply.java
```

Step 7: Create JAR File

Command:

```
jar cf multiply.jar Multiply*.class
```

Step 8: Verify All Files

Command:

```
ls ~/matrixmultiply/
```

```
athy@HP:~/matrixmultiply$ ls ~/matrixmultiply/  
M.txt          Multiply.class  input.txt  
'Multiply$MatrixMapper.class'  Multiply.java   multiply.jar  
'Multiply$MatrixReducer.class'  N.txt
```

Step 9: Run the MapReduce Job

Command:

```
hadoop jar multiply.jar Multiply ~/matrixmultiply/input.txt ~/matrixmultiply/output
```

Step 10: View the Output

Command:

```
hadoop fs -cat ~/matrixmultiply/output/part-r-00000
```

```
athy@HP:~/matrixmultiply$ hadoop fs -cat ~/matrixmultiply/output/part-r-00000  
0,0      58.0  
0,1      64.0  
1,0      139.0  
1,1      154.0
```

Name: Sabhya Lokhande

Rollno: BEAD-64

BDA Practical 4

Problem Statement: Develop a MapReduce program to find the grades of students using user input.

Create Program Files:

CreateInputFile.java

```
import java.io.FileWriter;
```

```
import java.util.Scanner;
```

```
public class CreateInputFile {  
    public static void main(String[] args) throws Exception {  
  
        Scanner sc = new Scanner(System.in);  
        FileWriter fw = new FileWriter("student.txt");  
  
        System.out.println("Enter number of students:");  
        int n = sc.nextInt();  
  
        for (int i = 0; i < n; i++) {  
            System.out.println("Enter name and marks:");  
            String name = sc.next();  
            int marks = sc.nextInt();  
  
            fw.write(name + " " + marks + "\n");  
        }  
        fw.close();  
        System.out.println("Input file created successfully!");  
    }  
}
```

GradeMapper.java

```
import java.io.IOException;
```

```
import org.apache.hadoop.io.*;
```

```
import org.apache.hadoop.mapreduce.Mapper;
```

```
public class GradeMapper extends Mapper<LongWritable, Text, Text, Text> {  
    public void map(LongWritable key, Text value, Context context) throws IOException,  
        InterruptedException {  
  
        String[] data = value.toString().split(" ");  
  
        String name = data[0];  
        int marks = Integer.parseInt(data[1]);  
        String grade;
```

```

        if (marks >= 90)
            grade = "A";
        else if (marks >= 75)
            grade = "B";
        else if (marks >= 50)
            grade = "C";
        else
            grade = "D";

        context.write(new Text(name), new Text(grade));
    }
}

```

GradeReducer.java

```

import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Reducer;

public class GradeReducer extends Reducer<Text, Text, Text, Text> {
    public void reduce(Text key, Iterable<Text> values, Context context) throws IOException,
        InterruptedException {
        for (Text val : values) {
            context.write(key, val);
        }
    }
}

```

GradeDriver.java

```

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class GradeDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.out.println("Usage: GradeDriver <input path> <output path>");
            System.exit(-1);
        }

        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Student Grades");
    }
}

```

```

        job.setJarByClass(GradeDriver.class);
        job.setMapperClass(GradeMapper.class);
        job.setReducerClass(GradeReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(Text.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}

```

Step 1: Compile and Run Input Program

Commands:

```
javac CreateInputFile.java
```

```
java CreateInputFile
```

```

vboxuser@Ubuntu-24:~/Desktop/BD4$ java CreateInputFile
Enter number of students:
4
Enter name and marks:
Stud1 88
Enter name and marks:
Stud2 95
Enter name and marks:
Stud3 72
Enter name and marks:
Stud4 50
Input file created successfully!

```

Step 2: Start Hadoop Services

Command:

```
start-dfs.sh
```

```
start-yarn.sh
```

```
jps
```

```

vboxuser@Ubuntu-24:~/Desktop/BD4$ start-dfs.sh
start-yarn.sh
jps
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Ubuntu-24]
Starting resourcemanager
Starting nodemanagers
4243 SecondaryNameNode
4468 ResourceManager
4602 NodeManager
4746 Jps
3871 NameNode
4015 DataNode
vboxuser@Ubuntu-24:~/Desktop/BD4$

```

Step 3: Compile MapReduce Program

Command:

```
javac -source 1.8 -target 1.8 -classpath `hadoop classpath` -d . *.java
```

```
vboxuser@Ubuntu-24:~/Desktop/BDA4$ javac -source 1.8 -target 1.8 -classpath `hadoop classpath` -d . *.java
warning: [options] bootstrap class path not set in conjunction with -source 8
warning: [options] source value 8 is obsolete and will be removed in a future release
warning: [options] target value 8 is obsolete and will be removed in a future release
warning: [options] To suppress warnings about obsolete options, use -Xlint:-options.
4 warnings
```

Step 4: Create JAR File

Command:

```
jar -cvf grades.jar *.class
```

```
vboxuser@Ubuntu-24:~/Desktop/BDA4$ jar -cvf grades.jar *.class
added manifest
adding: CreateInputFile.class(in = 1212) (out= 731)(deflated 39%)
adding: GradeDriver.class(in = 1485) (out= 838)(deflated 43%)
adding: GradeMapper.class(in = 1757) (out= 739)(deflated 57%)
adding: GradeReducer.class(in = 1488) (out= 622)(deflated 58%)
```

Step 5: Upload Input File to HDFS

Command:

```
hdfs dfs -rm -r input
```

```
hdfs dfs -mkdir input
```

```
hdfs dfs -put student.txt input/
```

```
vboxuser@Ubuntu-24:~/Desktop/BDA4$ hdfs dfs -rm -r input
Deleted input
vboxuser@Ubuntu-24:~/Desktop/BDA4$ hdfs dfs -mkdir input
vboxuser@Ubuntu-24:~/Desktop/BDA4$ hdfs dfs -put student.txt input/
vboxuser@Ubuntu-24:~/Desktop/BDA4$
```

Step 6: Execute MapReduce Program

Command:

```
hdfs dfs -rm -r output
```

```
hadoop jar grades.jar GradeDriver input output
```

```
vboxuser@Ubuntu-24:~/Desktop/BDA4$ hdfs dfs -rm -r output
Deleted output
vboxuser@Ubuntu-24:~/Desktop/BDA4$ hadoop jar grades.jar GradeDriver input output
2026-04-10 10:22:44,879 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-10 10:22:45,043 INFO input.FileInputFormat: Total input files to process : 1
2026-04-10 10:22:45,065 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-10 10:22:45,187 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1207908461_0001
2026-04-10 10:22:45,189 INFO mapreduce.JobSubmitter: Executing with tokens: []
```

Step 7: Display Output

Command:

```
hdfs dfs -cat output/part-r-00000
```

```
vboxuser@Ubuntu-24:~/Desktop/BDA4$ hdfs dfs -cat output/part-r-00000
Stud1    B
Stud2    A
Stud3    C
Stud4    C
vboxuser@Ubuntu-24:~/Desktop/BDA4$
```

Name: Sabhya Lokhande

Rollno: BEAD-64

BDA Practical 5

Problem Statement: Develop a MapReduce program to analyze Titanic ship data and to find the average age of the people (both male and female) who died in the tragedy. How many persons are survived in each class.

Create Program Files:

1. Average age of people who died (by gender)

AvgAgeMapper.java

```
import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Mapper;

public class AvgAgeMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    private Text gender = new Text();
    private IntWritable age = new IntWritable();

    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        if (line.startsWith("PassengerId")) return;

        String[] str = line.split(",(?=(?:[^\"]*"["\\"]*"\\\\"*)*["\\"]*$)");

        if (str.length > 5 && str[1].equals("1")) {

            if (!str[5].isEmpty()) {
                try {
                    int ageVal = (int) Float.parseFloat(str[5]);

                    gender.set(str[4]);
                    age.set(ageVal);

                    context.write(gender, age);
                } catch (Exception e) {}
            }
        }
    }
}
```

AvgAgeReducer.java

```

import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Reducer;

public class AvgAgeReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        int count = 0;

        for (IntWritable val : values) {
            sum += val.get();
            count++;
        }

        context.write(key, new IntWritable(sum / count));
    }
}

```

AvgAgeDriver.java

```

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AvgAgeDriver {

    public static void main(String[] args) throws Exception {

        Configuration conf = new Configuration();

        Job job = Job.getInstance(conf, "Average Age");

        job.setJarByClass(AvgAgeDriver.class);

        job.setMapperClass(AvgAgeMapper.class);
        job.setReducerClass(AvgAgeReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
    }
}

```

```

FileInputFormat.addInputPath(job, new Path(args[0]));
FileOutputFormat.setOutputPath(job, new Path(args[1]));

System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

2. Count of survivors per class

SurvivorMapper.java

```

import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Mapper;

public class SurvivorMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    private Text pclass = new Text();
    private final static IntWritable one = new IntWritable(1);

    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        if (line.startsWith("PassengerId")) return;

        String[] str = line.split("(?=(?:[^\"]*"\"[^\"]*"\"?)[^\"]*$)");

        if (str.length > 2 && str[1].equals("0")) {

            pclass.set("Class_" + str[2]);
            context.write(pclass, one);
        }
    }
}

```

SurvivorReducer.java

```

import java.io.IOException;

import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Reducer;

public class SurvivorReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

```

```

    int sum = 0;

    for (IntWritable val : values) {
        sum += val.get();
    }

    context.write(key, new IntWritable(sum));
}
}

```

SurvivorDriver.java

```

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class SurvivorDriver {

    public static void main(String[] args) throws Exception {

        Configuration conf = new Configuration();

        Job job = Job.getInstance(conf, "Survivors Per Class");

        job.setJarByClass(SurvivorDriver.class);

        job.setMapperClass(SurvivorMapper.class);
        job.setReducerClass(SurvivorReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}

```

Step 1: Start Hadoop

Commands:

```

start-dfs.sh
start-yarn.sh

```

```
vboxuser@Ubuntu-24:~/Desktop/BDAS$ start-dfs.sh
start-yarn.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Ubuntu-24]
Starting resourcemanager
Starting nodemanagers
```

Step 2: Exit Safe Mode

Command:

```
hdfs dfsadmin -safemode leave
```

```
vboxuser@Ubuntu-24:~/Desktop/BDAS$ hdfs dfsadmin -safemode leave
Safe mode is OFF
```

Step 3: Upload Dataset

Command:

```
hadoop fs -put Titanic.csv /
```

```
vboxuser@Ubuntu-24:~/Desktop/BDAS$ hadoop fs -put Titanic.csv /
```

Step 4: Compile Code

Command:

```
javac -classpath `hadoop classpath` -d . *.java
```

```
vboxuser@Ubuntu-24:~/Desktop/BDAS$ javac -classpath `hadoop classpath` -d . *.java
```

Step 5: Create JAR

Command:

```
jar -cvf titanic.jar *
```

```
vboxuser@Ubuntu-24:~/Desktop/BDAS$ jar -cvf titanic.jar *
added manifest
adding: AvgAgeDriver.class(in = 1363) (out= 752)(deflated 44%)
adding: AvgAgeDriver.java(in = 913) (out= 373)(deflated 59%)
adding: AvgAgeMapper.class(in = 2165) (out= 945)(deflated 56%)
adding: AvgAgeMapper.java(in = 942) (out= 443)(deflated 52%)
adding: AvgAgeReducer.class(in = 1614) (out= 682)(deflated 57%)
adding: AvgAgeReducer.java(in = 539) (out= 269)(deflated 50%)
adding: SurvivorDriver.class(in = 1379) (out= 754)(deflated 45%)
adding: SurvivorDriver.java(in = 929) (out= 374)(deflated 59%)
adding: SurvivorMapper.class(in = 2130) (out= 904)(deflated 57%)
adding: SurvivorMapper.java(in = 735) (out= 387)(deflated 47%)
adding: SurvivorReducer.class(in = 1600) (out= 670)(deflated 58%)
adding: SurvivorReducer.java(in = 489) (out= 251)(deflated 48%)
adding: Titanic.csv(in = 61113) (out= 21902)(deflated 64%)
```

Delete Old Outputs

```
hadoop fs -rm -r /output1
```

```
hadoop fs -rm -r /output2
```

Step 6: Run Job 1

Command:

```
hadoop jar titanic.jar AvgAgeDriver /Titanic.csv /output1
```

```
vboxuser@Ubuntu-24:~/Desktop/BDAS$ hadoop jar titanic.jar AvgAgeDriver /Titanic.csv /output1
2026-04-11 08:27:59,974 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-11 08:28:00,093 INFO input.FileInputFormat: Total input files to process : 1
2026-04-11 08:28:00,187 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-11 08:28:00,346 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local553516831_0001
2026-04-11 08:28:00,347 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-11 08:28:00,515 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-11 08:28:00,516 INFO mapreduce.Job: Running job: job_local553516831_0001
2026-04-11 08:28:00,518 INFO mapred.LocalJobRunner: OutputCommitter set in config null
```

View Output 1

`hadoop fs -cat /output1/part-r-00000`

```
vboxuser@Ubuntu-24:~/Desktop/BDAS$ hadoop fs -cat /output1/part-r-00000
female 28
male 27
```

Step 7: Run Job 2

Command:

`hadoop jar titanic.jar SurvivorDriver /Titanic.csv /output2`

```
vboxuser@Ubuntu-24:~/Desktop/BDAS$ hadoop jar titanic.jar SurvivorDriver /Titanic.csv /output2
2026-04-11 08:28:21,901 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-11 08:28:22,051 INFO input.FileInputFormat: Total input files to process : 1
2026-04-11 08:28:22,141 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-11 08:28:22,365 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local475408257_0001
2026-04-11 08:28:22,367 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-11 08:28:22,551 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-11 08:28:22,554 INFO mapreduce.Job: Running job: job_local475408257_0001
2026-04-11 08:28:22,556 INFO mapred.LocalJobRunner: OutputCommitter set in config null
```

View Output 2

`hadoop fs -cat /output2/part-r-00000`

```
vboxuser@Ubuntu-24:~/Desktop/BDAS$ hadoop fs -cat /output2/part-r-00000
Class_1 80
Class_2 97
Class_3 372
vboxuser@Ubuntu-24:~/Desktop/BDAS$
```

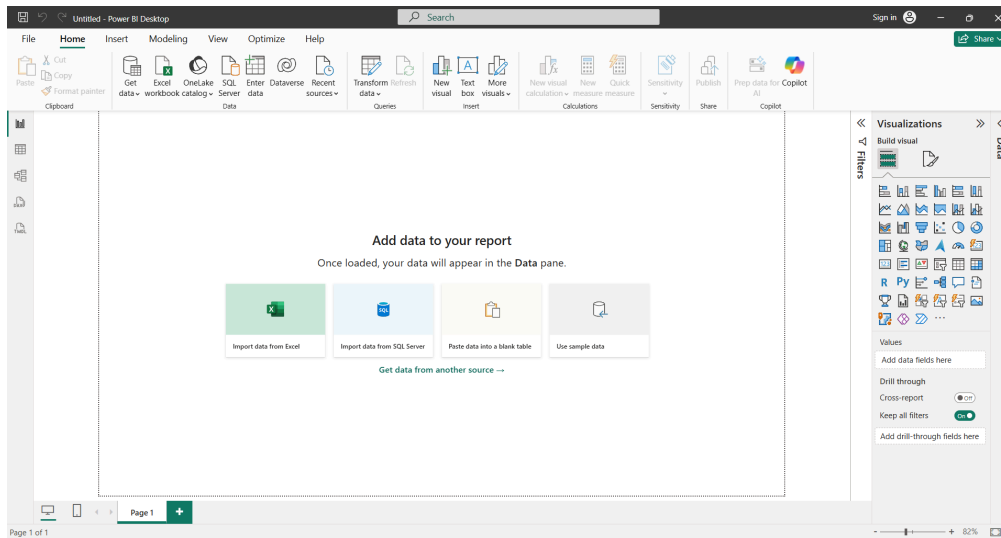

Name: Sabhya Lokhande

Rollno: BEAD-64

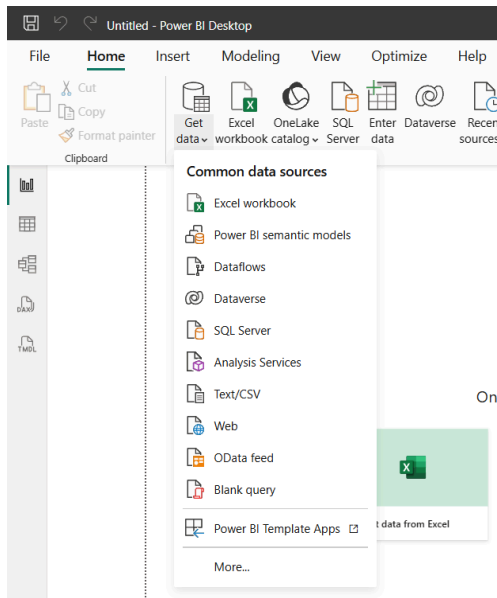
BI Practical 6

Problem Statement: Import Data from different Sources such as (Excel, Sql Server, Oracle etc.) and load in targeted system.

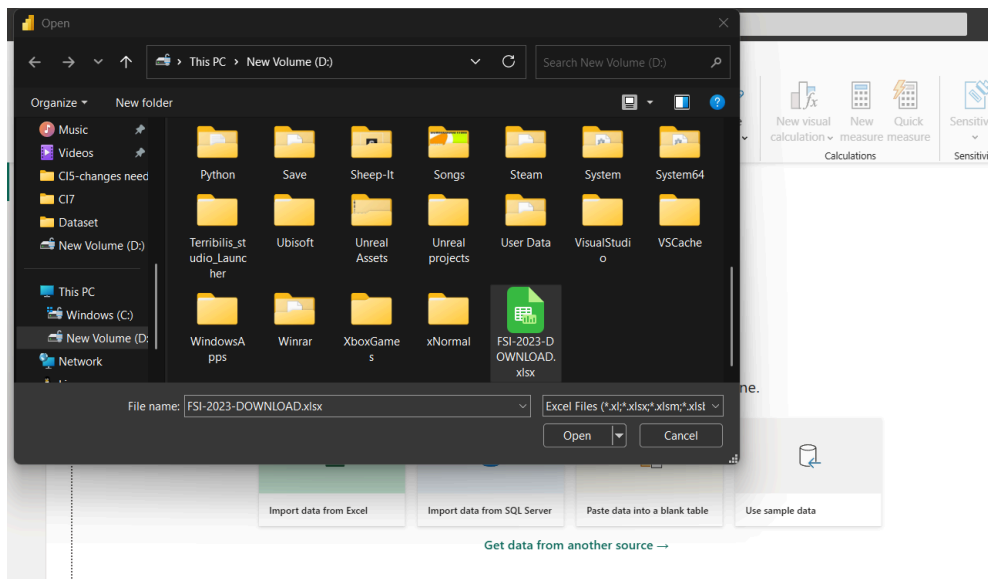
1) Launch Power BI Desktop.



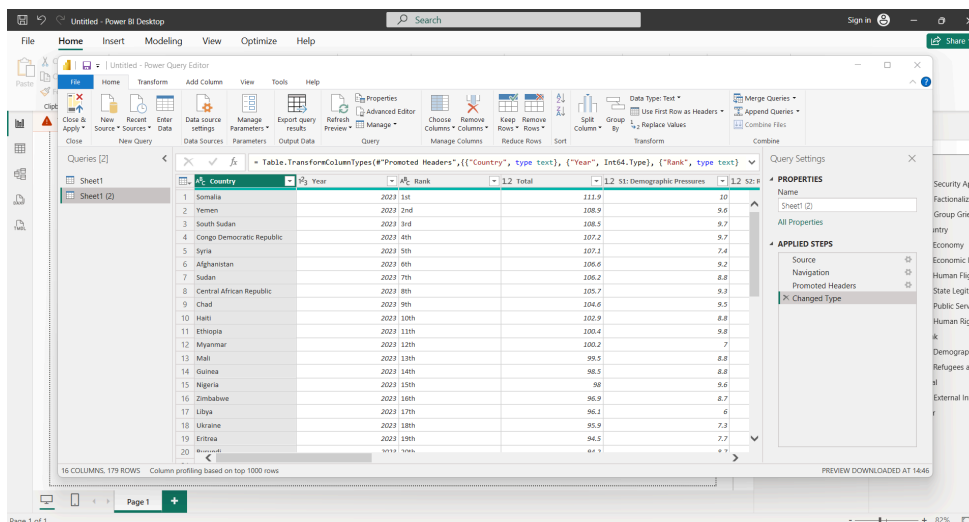
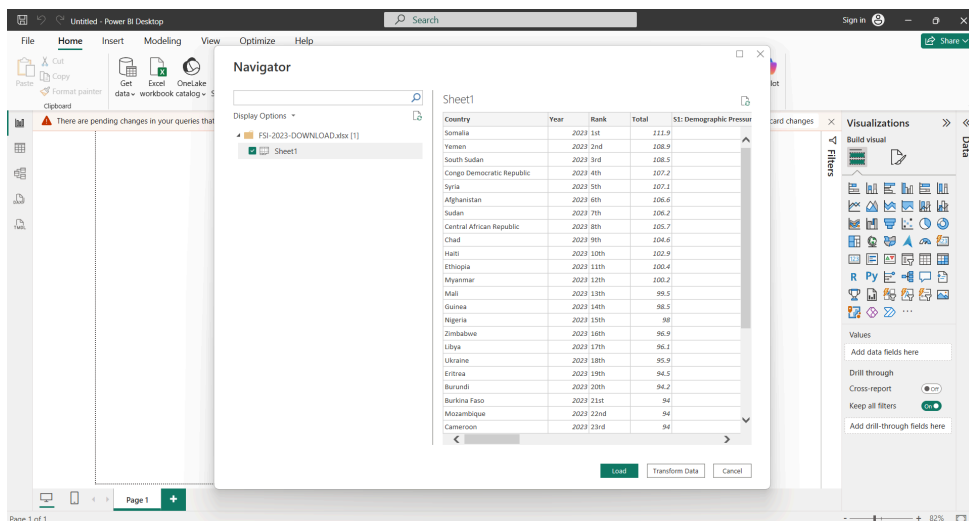
2) From the Home ribbon, select Get Data.



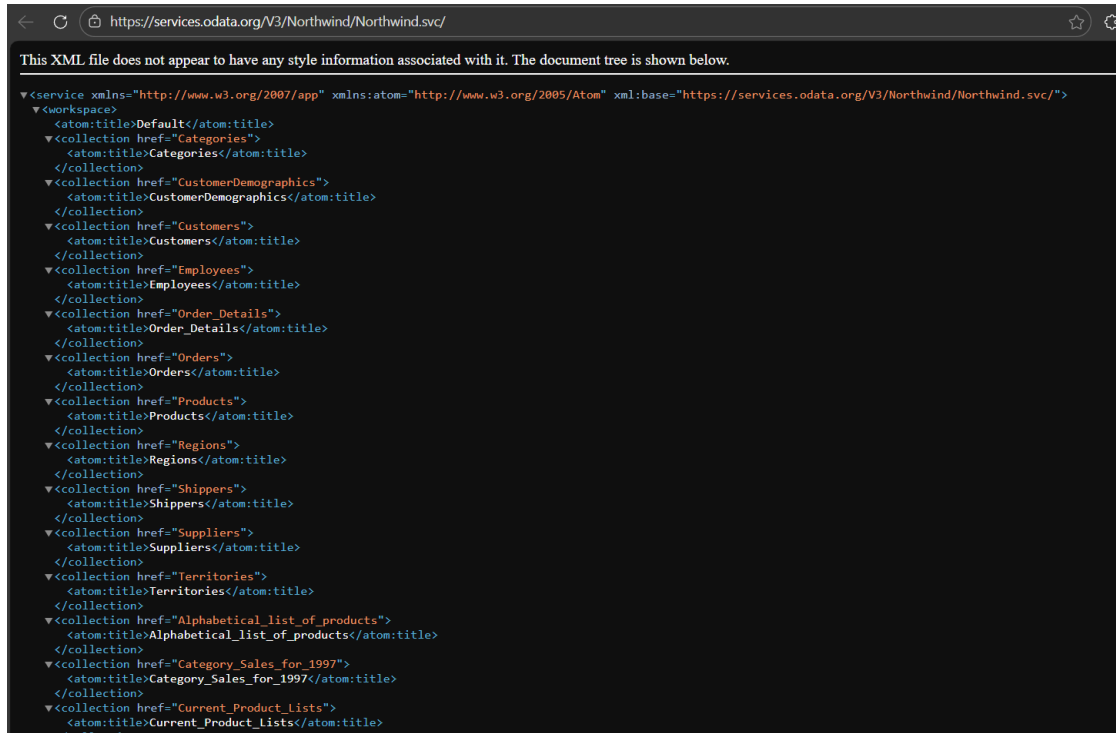
3) In the Open File dialog box, select the .xlsx file.



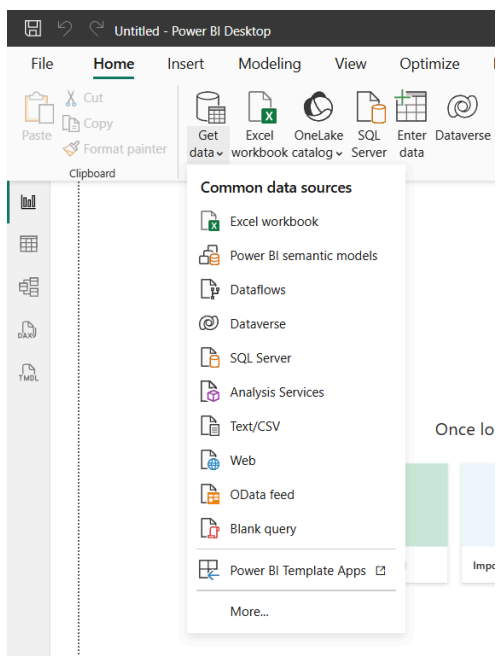
4) In the Navigator pane, select the Products table and then select Transform Data to make changes if any or select Load to load file in your report.



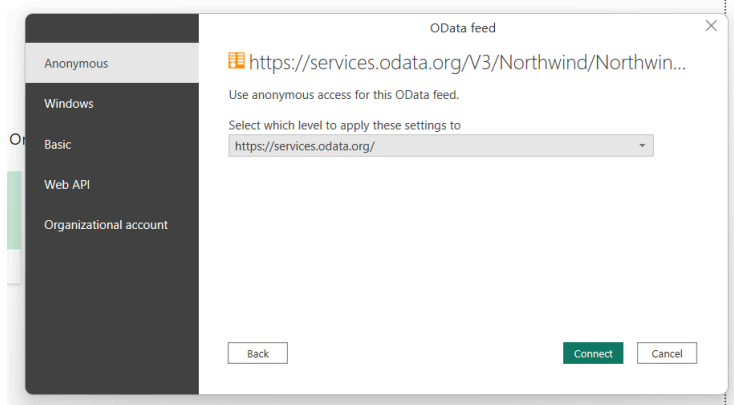
Importing Data from OData Feed: (<http://services.odata.org/V3/Northwind/Northwind.svc/>)



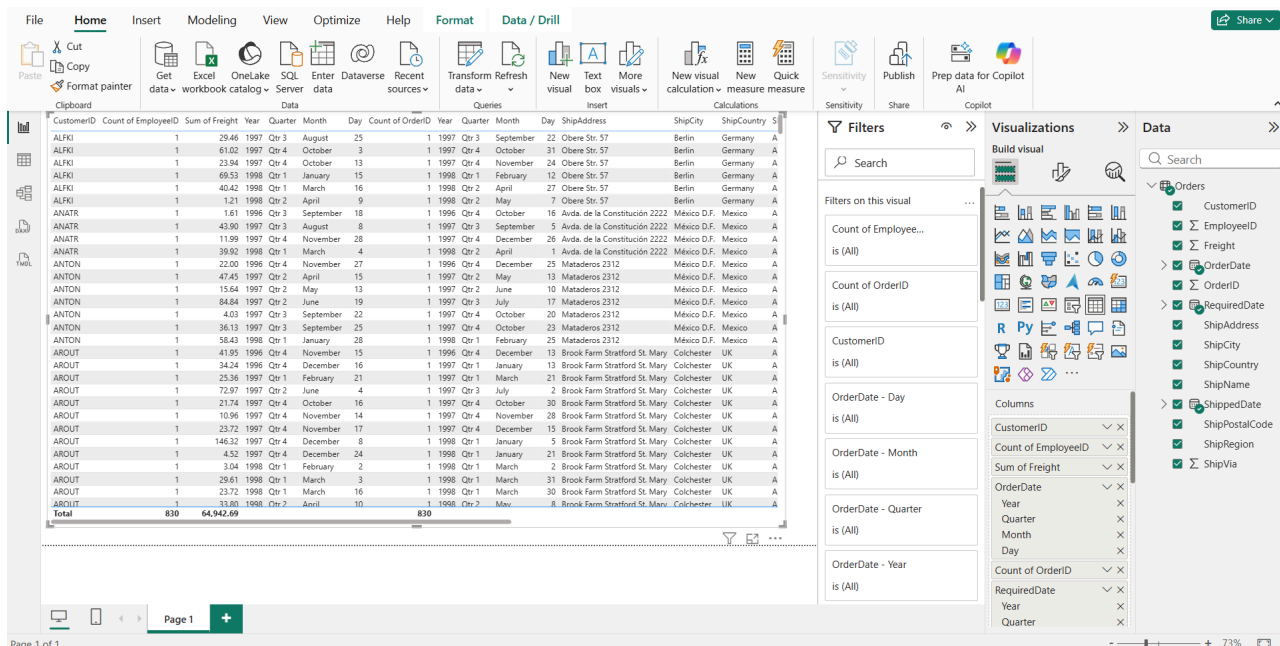
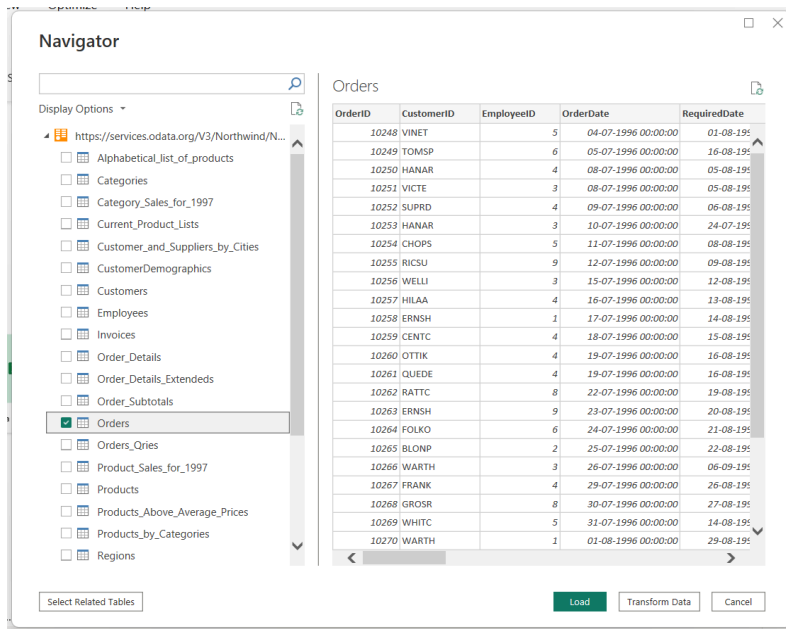
1) From the Home ribbon tab in the Query Editor, select Get Data. Browse to the OData Feed data source.



2) In the OData Feed dialog box, paste the URL for the Northwind OData feed.



3) In the Navigator pane, select the Orders table, and then select Edit.



Name: Sabhya Lokhande

Rollno: BEAD-64

BI Practical 7

Problem Statement: Data Visualization from Extraction Transformation and Loading (ETL) Process.

Step 1: Check Data (Extraction)

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("D:/College/BE/Dataset/Iris.csv")

print(df.head())
```

	SepallLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Step 2: Transform

Check missing values

```
print(df.isnull().sum())
```

```
SepallLengthCm    0
SepalWidthCm      0
PetalLengthCm     0
PetalWidthCm      0
Species           0
dtype: int64
```

Create new column

```
df['PetalArea'] = df['PetalLengthCm'] * df['PetalWidthCm']
```

Convert species names

```
df['Species'] = df['Species'].str.replace('Iris-', '')
```

```
df.head()
```

	SepallLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	PetalArea
0	5.1	3.5	1.4	0.2	setosa	0.28
1	4.9	3.0	1.4	0.2	setosa	0.28
2	4.7	3.2	1.3	0.2	setosa	0.26
3	4.6	3.1	1.5	0.2	setosa	0.30
4	5.0	3.6	1.4	0.2	setosa	0.28

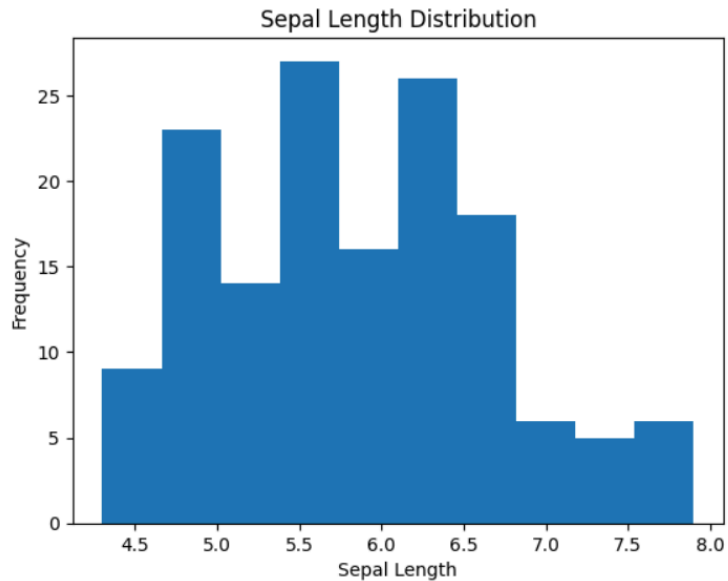
Step 3: Load

```
df.to_csv("D:/College/BE/Dataset/iris_cleaned.csv", index=False)
```

Step 4: Visualization

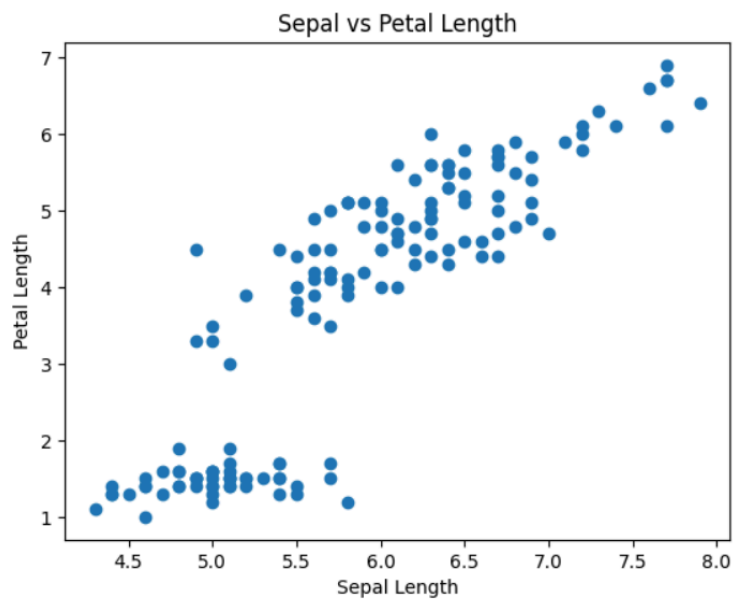
Sepal Length Distribution

```
plt.hist(df['SepalLengthCm'])  
plt.title("Sepal Length Distribution")  
plt.xlabel("Sepal Length")  
plt.ylabel("Frequency")  
plt.show()
```



Scatter Plot

```
plt.scatter(df['SepalLengthCm'], df['PetalLengthCm'])  
plt.title("Sepal vs Petal Length")  
plt.xlabel("Sepal Length")  
plt.ylabel("Petal Length")  
plt.show()
```



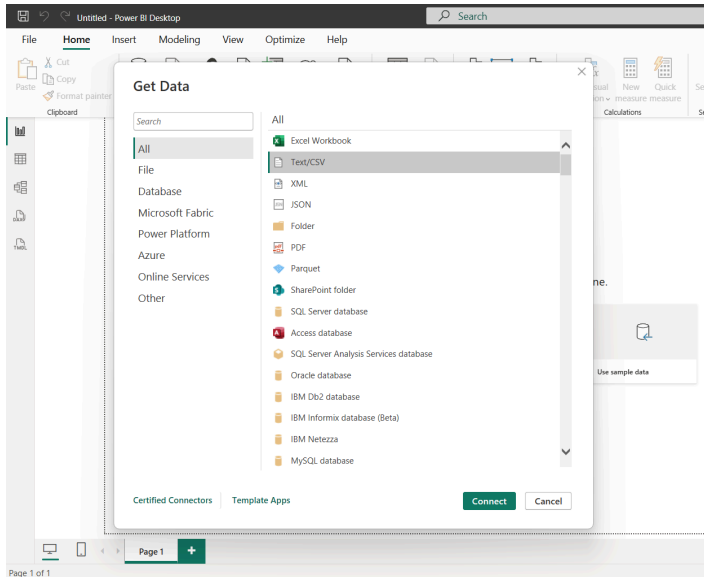
Name: Sabhya Lokhande

Rollno: BEAD-64

BI Practical 8

Problem Statement: Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

STEP 1: Extraction (Load Data)



Titanic-Dataset.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450
6	0	3	Moran, Mr. James	male	null	0	0	330877
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736
11	1	3	Sandstrom, Miss. Marguerite Rut	female	4	1	1	PP 9549
12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783
13	0	3	Saunderscock, Mr. William Henry	male	20	0	0	A/5. 2151
14	0	3	Andersson, Mr. Anders Johan	male	39	1	5	347082
15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14	0	0	350406
16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55	0	0	248706
17	0	3	Rice, Master. Eugene	male	2	4	1	382652
18	1	2	Williams, Mr. Charles Eugene	male	null	0	0	244373
19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vandemoortel...)	female	31	1	0	345763
20	1	3	Massei, Mrs. Fatima	female	null	0	0	2649

Extract Table Using Examples

Load Transform Data Cancel

STEP 2: Transformation

(A) Remove Unnecessary Columns (Select the ones you want to keep)

Query Editor

Add Column View Tools Help

Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Query Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Any Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Remove Other Columns

Table.TransformColumnTypes(#Promoted Headers,{{"PassengerId", Int64.Type}, {"Survived", Int64.Type}, {"Pclass", Int64.Type}, {"Name",

	1.2 SibSp	1.2 Parch	1.2 Ticket	1.2 Fare	1.2 Cabin	1.2 Embarked
1	22	1	0 A/5 21171	7.25		S
2	38	1	0 PC 17599	71.2833	C85	C
3	26	0	0 STON/O2. 3101282	7.925		S
4	35	1	0 113803	53.1	C123	S
5	35	0	0 373450	8.05		S
6	null	0	0 330877	8.4583		Q
7	54	0	0 17463	51.8625	F46	S

(B) Handle Missing Values (Replace with Average)

Tools Help

Replace Values Unpivot Columns Merge Columns

Fill Move Split Column Format Extract Parse

Any Column Text Column

Statistics Standard Scientific Rounding Information

Sum Minimum Maximum Median Average Standard Deviation Count Values Count Distinct Values

table.SelectColumns(#"Changed Type",{"PassengerId", "Survived", "Pclass", "Age", "SibSp", "Parch",

1.2 Age	1.2 SibSp	1.2 Parch
22	1	
38	1	
26	0	
35	1	

fx = List.Average(#"Removed Other Columns"[Age])

29.699117647058824

1.2 PassengerId	1.2 Survived	1.2 Pclass	1.2 Sex	1.2 Age	1.2 SibSp
1	0	3	male	22	
2	1	1	female	38	
3	0	3	female	26	
4	1	1	female	35	
5	0	3	male	35	
6	1	3	female	null	
7	0	1	male	54	
8	1	3	female	2	
9	0	1	male	27	
10	1	3	female	14	
11	0	1	male	4	
12	1	3	female	58	
13	0	3	male	20	
14	1	1	female	39	
15	0	1	male	14	
16	1	3	female	55	
17	0	1	male	2	
18	1	1	female	51	
19	0	3	male	52	
20	1	1	female	35	
21	0	3	male	51	
22	1	1	female	35	
23	0	3	male	47	
24	1	1	female	35	
25	0	3	male	47	
26	1	1	female	35	
27	0	3	male	47	
28	1	1	female	35	
29	0	3	male	47	
30	1	1	female	35	

Replace Values

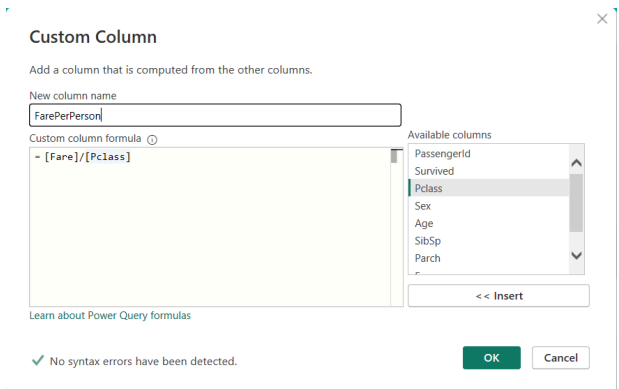
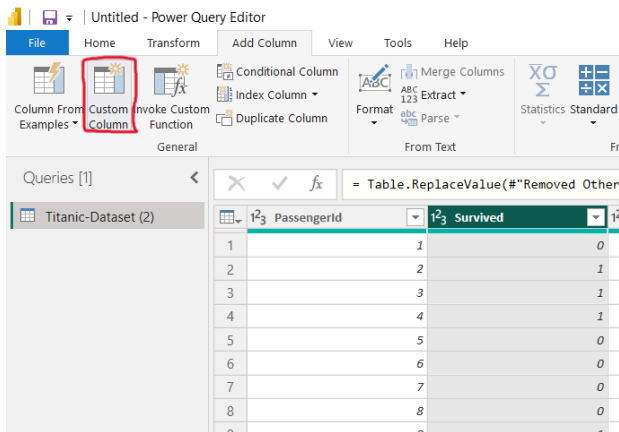
Replace one value with another in the selected columns.

Value To Find: null

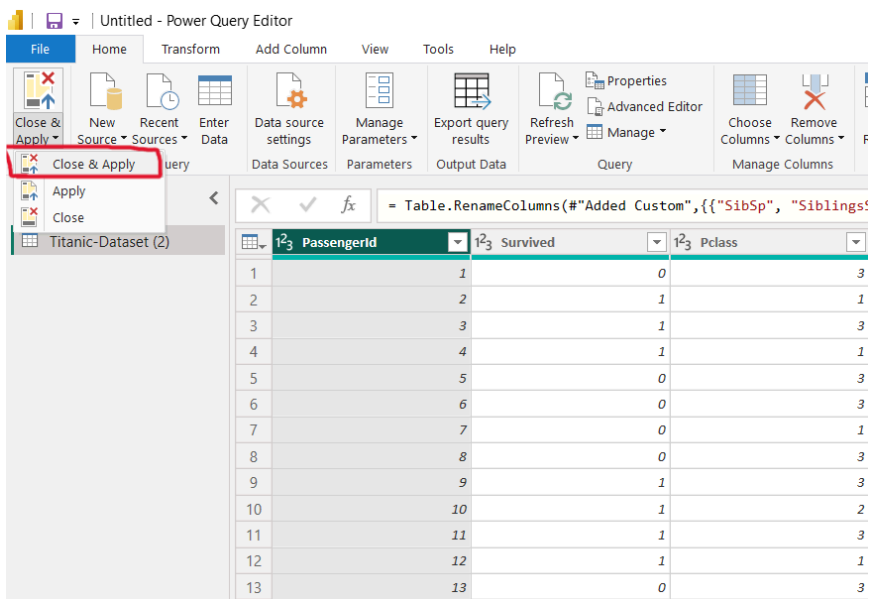
Replace With: 29

OK Cancel

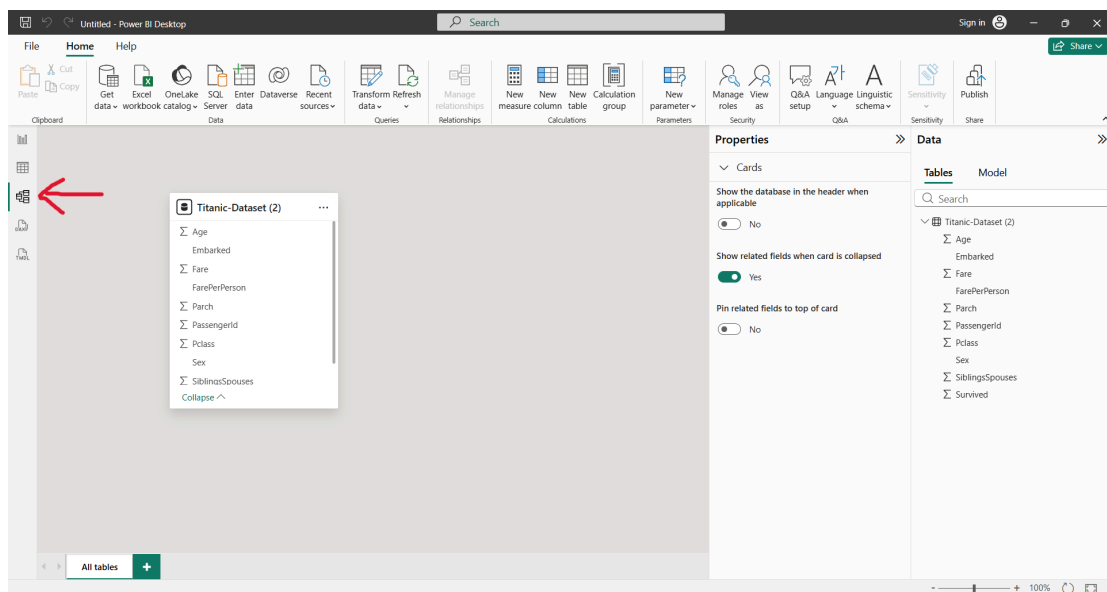
(C) Create NEW Column



STEP 3: Loading



STEP 4: Database Construction



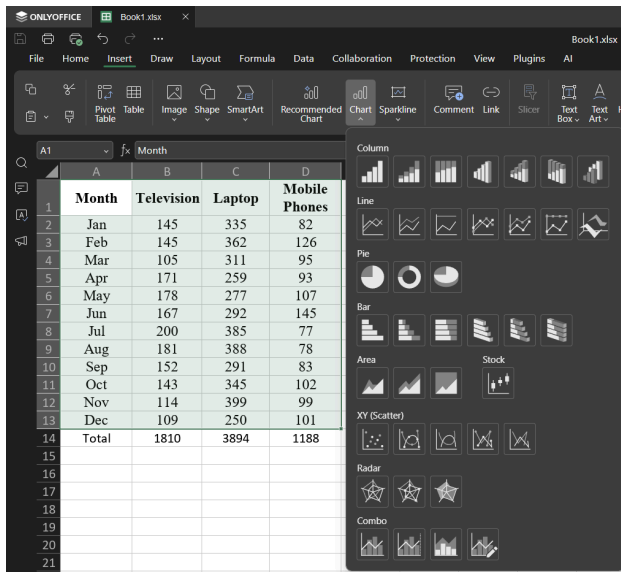
Name: Sabhya Lokhande

Rollno: BEAD-64

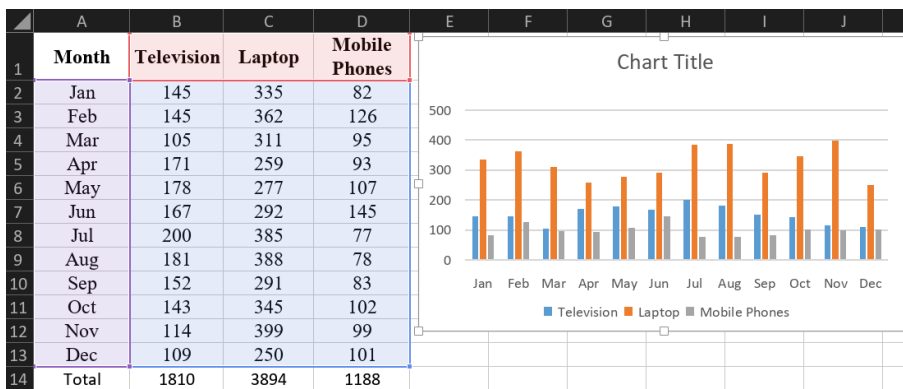
BI Practical 9

Problem Statement: Data Analysis and Visualization using Advanced Excel.

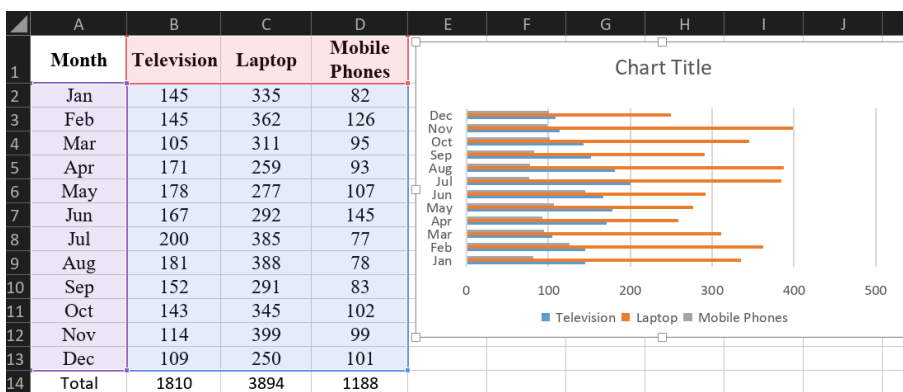
STEP 1: Open Excel & Enter Data



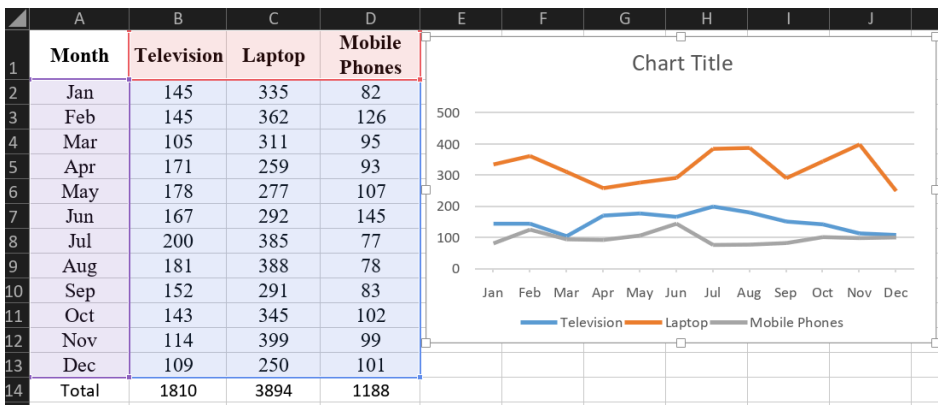
STEP 2: Create COLUMN CHART



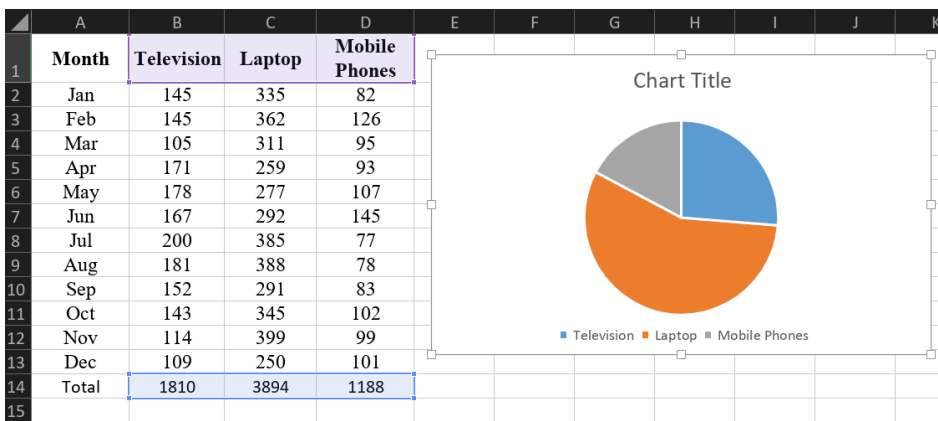
STEP 3: Create BAR CHART



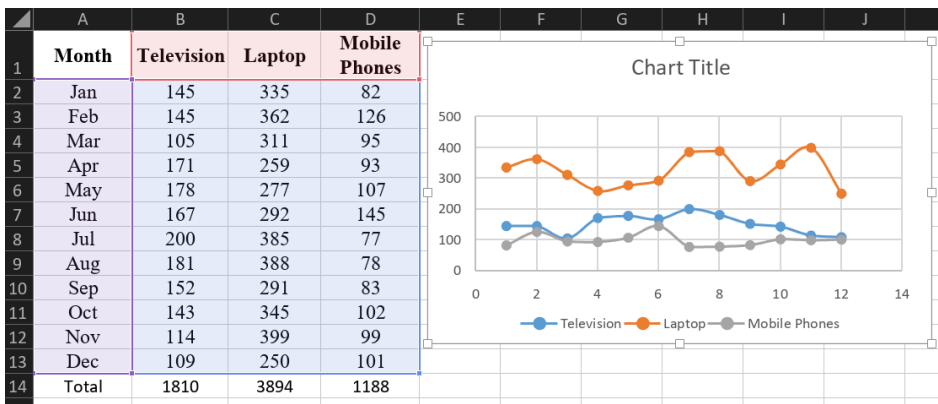
STEP 4: Create LINE CHART



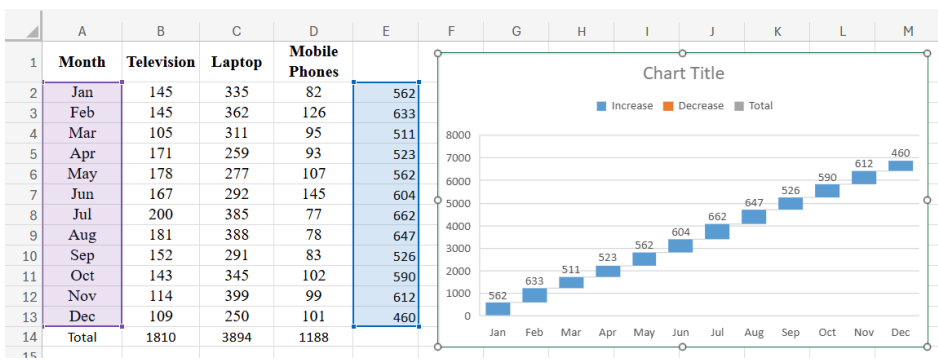
STEP 5: Create PIE CHART



STEP 6: Create SCATTER PLOT

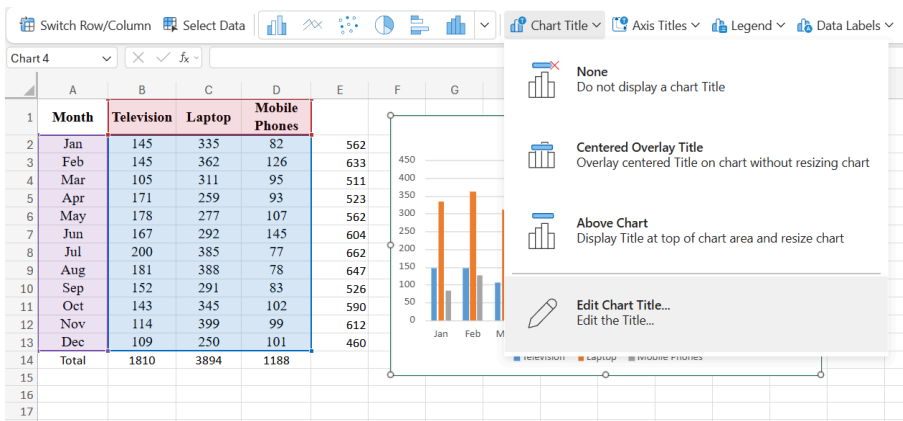


STEP 7: Create WATERFALL CHART



STEP 8: Customize Chart

Edit Chart Title



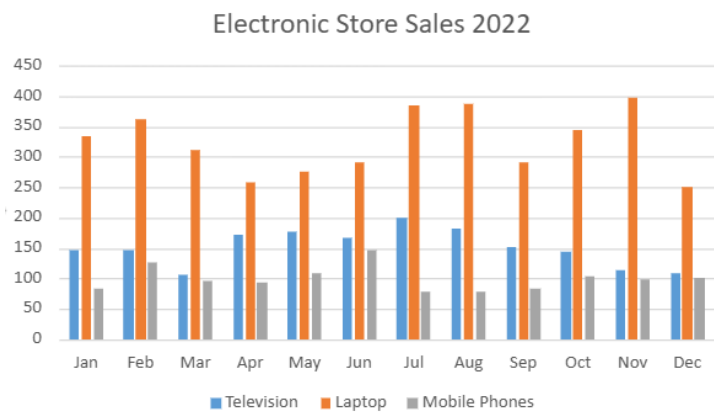
Edit Title

Title text:

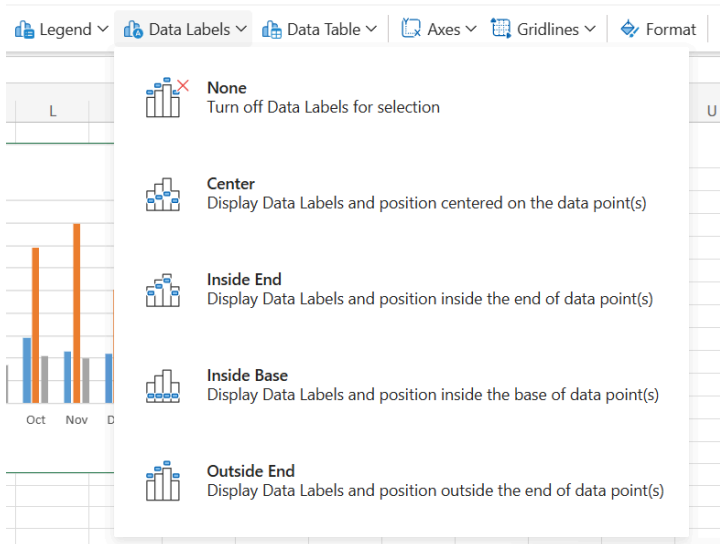
Electronic Store Sales 2022

OK

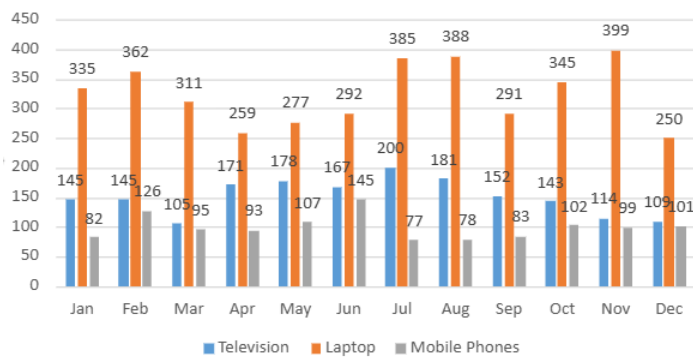
Cancel



Display Data Labels



Electronic Store Sales 2022



STEP 5: Split Data and Apply Classification Algorithm

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3)
```

```
model = LogisticRegression(max_iter=200)  
model.fit(X_train, y_train)
```

▼ LogisticRegression ⓘ ?

► Parameters

STEP 6: Prediction

```
y_pred = model.predict(X_test)
```

STEP 7: Evaluation

```
cm = confusion_matrix(y_test, y_pred)  
disp = ConfusionMatrixDisplay(confusion_matrix=cm)  
disp.plot(cmap='Blues')  
plt.show()
```

